



COMPUTER
ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	10/08/2026	DATE SUBMITTED:	10/08/2026
NAME:	Interno, Jerald San M.	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 11.1 Containerization

1. PROCEDURE OUTPUT

What procedures did you perform?

Task
1. Create a new repository for this activity. 2. Install Docker and enable the docker socket. 3. Add to Docker group to your current user. 4. Create a Dockerfile to install web and DB server. 5. Install and build the Dockerfile using Ansible. 6. Add, commit and push it to your repository.
Screenshot
Install Docker and enable the docker socket.



```
(base) jerald@workstation:~$ sudo apt update
[sudo] password for jerald:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,375 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,073 kB]
Get:7 http://ph.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [306 kB]
Get:8 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:9 http://ph.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,730 kB]
Get:10 http://ph.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [396 kB]
Get:11 http://ph.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,702 kB]
Get:12 http://ph.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [341 kB]
Get:13 http://ph.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:14 http://ph.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [13.3 kB]
Get:15 http://ph.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:16 http://ph.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5,752 B]
Get:17 http://ph.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [223 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1,566 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [361 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,218 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [245 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [11.6 kB]
Fetched 11.6 MB in 16s (751 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
171 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
(base) jerald@workstation:~$ sudo apt install ca-certificates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20260601~24.04.1).
ca-certificates set to manually installed.
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64
The following packages will be upgraded:
  curl libcurl3t64-gnutls libcurl4t64
3 upgraded, 0 newly installed, 0 to remove and 168 not upgraded.
Need to get 906 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.1
227 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubu
10.15 [343 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5
2ubuntu10.15 [336 kB]
Fetched 906 kB in 2s (388 kB/s)
(Reading database ... 257976 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.15_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.15) over (8.5.0-2ubuntu10.13) ...
Preparing to unpack .../libcurl4t64_8.5.0-2ubuntu10.15_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.5.0-2ubuntu10.15) over (8.5.0-2ubuntu10.13) ...
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.15_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.15) over (8.5.0-2ubuntu10.13
) ...
Setting up libcurl4t64:amd64 (8.5.0-2ubuntu10.15) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.15) ...
Setting up curl (8.5.0-2ubuntu10.15) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
```

```
(base) jerald@workstation:~$ sudo install -m 0755 -d /etc/apt/keyrings
(base) jerald@workstation:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/
apt/keyrings/docker.asc
(base) jerald@workstation:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
```

```
(base) jerald@workstation:~$ sudo nano /etc/apt/sources.list.d/docker.sources
```

```
GNU nano 7.2 /etc/apt/sources.list.d/docker.sources
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: noble
Components: stable
Architectures: amd64
Signed-By: /etc/apt/keyrings/docker.asc
```

```
(base) jerald@workstation:~$ sudo apt update
[sudo] password for jerald:
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:2 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [68.2 kB]
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:7 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,375 kB]
Get:8 http://ph.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [306 kB]
Get:9 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:10 http://ph.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,761 kB]
Get:11 http://ph.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [401 kB]
Get:12 http://ph.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,702 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,106 kB]
Get:14 http://ph.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [341 kB]
Get:15 http://ph.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:16 http://ph.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [55.0 kB]
Get:17 http://ph.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [14.3 kB]
Get:18 http://ph.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [226 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.3 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1,613 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [371 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,218 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [245 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Fetched 11.8 MB in 10s (1,212 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
170 packages can be upgraded. Run 'apt list --upgradable' to see them.
```



```
(base) jerald@workstation:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras pigz
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras
  docker-compose-plugin pigz
0 upgraded, 7 newly installed, 0 to remove and 170 not upgraded.
Need to get 101 MB of archives.
After this operation, 386 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.3.6-1-ubuntu.24.04-noble [23.2 MB]
Get:2 http://ph.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:3 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:29.8.2-1-ubuntu.24.04-noble [17.5 MB]
Get:4 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:29.8.2-1-ubuntu.24.04-noble [24.4 MB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.38.0-1-ubuntu.24.04-noble [17.8 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:29.8.2-1-ubuntu.24.04-noble [10.2 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 5.6.0-1-ubuntu.24.04-noble [8,080 kB]
Fetched 101 MB in 19s (5,220 kB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 257976 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_2.3.6-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking containerd.io (2.3.6-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5%3a29.8.2-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-cli (5:29.8.2-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5%3a29.8.2-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce (5:29.8.2-1-ubuntu.24.04-noble) ...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.38.0-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-buildx-plugin (0.38.0-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5%3a29.8.2-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:29.8.2-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../6-docker-compose-plugin_5.6.0-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-compose-plugin (5.6.0-1-ubuntu.24.04-noble) ...
Setting up docker-buildx-plugin (0.38.0-1-ubuntu.24.04-noble) ...
Setting up containerd.io (2.3.6-1-ubuntu.24.04-noble) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/
```

```
(base) jerald@workstation:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-10-08 22:28:59 PST; 49s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 8478 (dockerd)
    Tasks: 12
   Memory: 30.3M (peak: 31.0M)
     CPU: 1.818s
    CGroup: /system.slice/docker.service
            └─8478 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 08 22:28:57 workstation dockerd[8478]: time="2026-10-08T22:28:57.381569045+08:00" level=
Oct 08 22:28:57 workstation dockerd[8478]: time="2026-10-08T22:28:57.447710762+08:00" level=
Oct 08 22:28:57 workstation dockerd[8478]: time="2026-10-08T22:28:57.476998795+08:00" level=
Oct 08 22:28:58 workstation dockerd[8478]: time="2026-10-08T22:28:58.654796146+08:00" level=
Oct 08 22:28:58 workstation dockerd[8478]: time="2026-10-08T22:28:58.703093005+08:00" level=
Oct 08 22:28:58 workstation dockerd[8478]: time="2026-10-08T22:28:58.703440482+08:00" level=
Oct 08 22:28:59 workstation dockerd[8478]: time="2026-10-08T22:28:59.167399534+08:00" level=
Oct 08 22:28:59 workstation dockerd[8478]: time="2026-10-08T22:28:59.174251746+08:00" level=
Oct 08 22:28:59 workstation dockerd[8478]: time="2026-10-08T22:28:59.174502645+08:00" level=
Oct 08 22:28:59 workstation systemd[1]: Started docker.service - Docker Application Containe
lines 1-22/22 (END)
```

Add to Docker group to your current user.

```
(base) jerald@workstation:~$ sudo usermod -aG docker $USER
(base) jerald@workstation:~$ newgrp docker
(base) jerald@workstation:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Create a Dockerfile to install web and DB server.

```
(base) jerald@workstation:~$ mkdir -p ~/docker-ansible-activity
(base) jerald@workstation:~$ cd ~/docker-ansible-activity
(base) jerald@workstation:~/docker-ansible-activity$ nano Dockerfile
```

```
GNU nano 7.2 Dockerfile *
FROM ubuntu:latest

#to avoid the interactive prompts during apt installation
ENV DEBIAN_FRONTEND=noninteractive

#installation of Apache2 web server and MariaDB database server
RUN apt-get update && apt-get install -y \
    apache2 \
    mariadb-server \
    && rm -rf /var/lib/apt/lists/*

#expose HTTP (80) and MySQL/MariaDB (3306) ports
EXPOSE 80 3306

#start the MariaDB service and run Apache in the foreground
CMD service mariadb start && apachectl -D FOREGROUND
```

Install and build the Dockerfile using Ansible.

```
(base) jerald@workstation:~/docker-ansible-activity$ pip install docker
Collecting docker
  Downloading docker-7.2.0-py3-none-any.whl.metadata (3.8 kB)
Requirement already satisfied: requests>=2.26.0 in /home/jerald/miniconda3/lib/python3.14/site-packages (from docker) (2.34.2)
Requirement already satisfied: urllib3>=1.26.0 in /home/jerald/miniconda3/lib/python3.14/site-packages (from docker) (2.7.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /home/jerald/miniconda3/lib/python3.14/site-packages (from requests>=2.26.0->docker) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /home/jerald/miniconda3/lib/python3.14/site-packages (from requests>=2.26.0->docker) (3.18)
Requirement already satisfied: certifi>=2023.5.7 in /home/jerald/miniconda3/lib/python3.14/site-packages (from requests>=2.26.0->docker) (2026.7.22)
Downloading docker-7.2.0-py3-none-any.whl (148 kB)
Installing collected packages: docker
Successfully installed docker-7.2.0
```

```
(base) jerald@workstation:~/docker-ansible-activity$ ansible-galaxy collection install community.docker
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to reinstall them, consider using '--force'.
(base) jerald@workstation:~/docker-ansible-activity$ nano site.yml
(base) jerald@workstation:~/docker-ansible-activity$ ansible-playbook site.yml
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit
localhost does not match 'all'

PLAY [Build Web and DB Server Docker Image] *****

TASK [Gathering Facts] *****
ok: [localhost]

TASK [Build Docker image from Dockerfile] *****
changed: [localhost]

PLAY RECAP *****
localhost                : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

GNU nano 7.2

site.yml *

```
---
- name: Build Web and DB Server Docker Image
  hosts: localhost
  connection: local
  tasks:
    - name: Build Docker image from Dockerfile
      community.docker.docker_image:
        name: web_db_server
        build:
          path: .
          source: build
        state: present
```

```
(base) jerald@workstation:~/docker-ansible-activity$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	5e2309035332	25.9kB	9.49kB	U
ubuntu:latest	f144425ff09b	162MB	45.6MB	
web_db_server:latest	27b860055d2d	625MB	112MB	

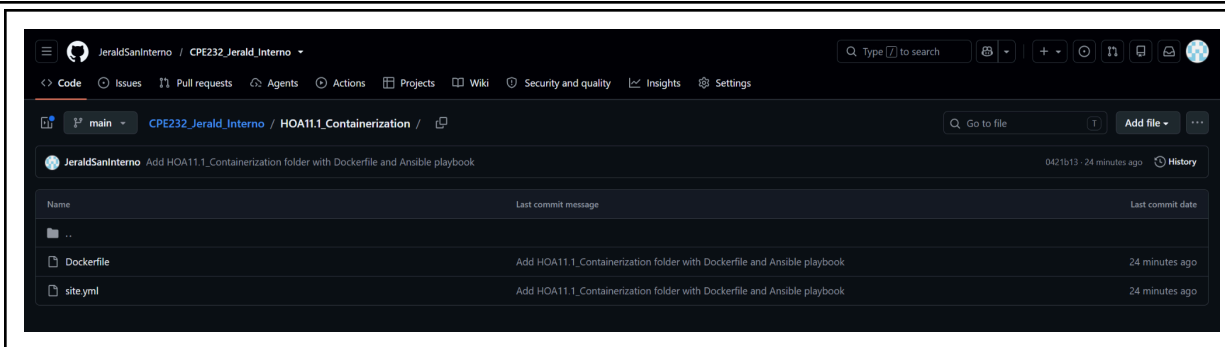
i Info → U In Use

Add, commit and push it to your repository.

```
(base) jerald@workstation:~/my-repo$ git init -b main
Initialized empty Git repository in /home/jerald/my-repo/.git/
(base) jerald@workstation:~/my-repo$ cd ~
(base) jerald@workstation:~$ git init -b main
Initialized empty Git repository in /home/jerald/.git/
(base) jerald@workstation:~$ cd my-repo
(base) jerald@workstation:~/my-repo$ mv ~/docker-ansible-activity ./HOA11.1_Containerization
(base) jerald@workstation:~/my-repo$ git add .
(base) jerald@workstation:~/my-repo$ git commit -m "Add HOA11.1_Containerization folder with Dockerfile and Ansible playbo
ok"
[main (root-commit) 0421b13] Add HOA11.1_Containerization folder with Dockerfile and Ansible playbook
 2 files changed, 28 insertions(+)
 create mode 100644 HOA11.1_Containerization/Dockerfile
 create mode 100644 HOA11.1_Containerization/site.yml
(base) jerald@workstation:~/my-repo$ git remote add origin https://github.com/JeraldSanInterno/CPE232_Jerald_Interno.git
```

```
(base) jerald@workstation:~/my-repo$ git pull origin main --allow-unrelated-histories --no-edit
From https://github.com/JeraldSanInterno/CPE232_Jerald_Interno
* branch      main      -> FETCH_HEAD
Merge made by the 'ort' strategy.
.../1.1 Hands-on Activity - Creating Virtual Machines .pdf | Bin 0 -> 2015277 bytes
HOA1.1_Creating_Virtual_Machines/README.md               | 1 +
...Configure and Manage Enterprise Log Monitoring via Ansible_Interno (1).pdf | Bin 0 -> 1080013 bytes
HOA10.1_Log_Monitoring/README.md                         | 1 +
.../Hands-on Activity_ Familiarization for the new normal Lab.pdf | Bin 0 -> 298361 bytes
HOA1_Familiarization_New_Normal/README.md               | 1 +
.../2.1 Hands-on Activity - SSH Key-Based Authentication and GIT Setup.pdf | Bin 0 -> 1072507 bytes
HOA2.1_SSH_Key_Auth_and_Git_Setup/README.md             | 1 +
.../3.1 Hands-on Activity - Install SSH server on CentOS or RHEL 8.pdf | Bin 0 -> 442265 bytes
HOA3.1_Install_SSH_Server_CentOS_RHEL8/README.md        | 1 +
HOA4.1_Ansible_Basics/Activity No. 4 Running Elevated Ad hoc Commands.pdf | Bin 0 -> 1643750 bytes
HOA4.1_Ansible_Basics/README.md                         | 1 +
HOA4.1_Ansible_Basics/install_apache.yml               | 15 +++++
HOA4.1_Ansible_Basics/inventory.yml                   | 6 ++
.../Hands-on Activity 5.1_ Implement Ansible roles in playbooks.pdf | Bin 0 -> 1265523 bytes
HOA5.1_Ansible_Roles_in_Playbooks/README.md            | 1 +
HOA5.1_Ansible_Roles_in_Playbooks/install_apache.yml   | 30 ++++++++
HOA5.1_Ansible_Roles_in_Playbooks/inventory.yml        | 28 ++++++++
.../Hands-On Activity 6.1 Targeting Specific Nodes_Interno.pdf | Bin 0 -> 3156773 bytes
HOA6.1_Targeting_Specific_Nodes/README.md              | 1 +
HOA6.1_Targeting_Specific_Nodes/inventory.yml           | 23 ++++++
HOA6.1_Targeting_Specific_Nodes/roles/.gitkeep         | 4 ++
HOA6.1_Targeting_Specific_Nodes/roles/dbserver/main.yml | 12 ++++
HOA6.1_Targeting_Specific_Nodes/roles/fileserver/main.yml | 5 ++
HOA6.1_Targeting_Specific_Nodes/roles/webserver/main.yml | 23 ++++++
HOA6.1_Targeting_Specific_Nodes/site.yml               | 93 ++++++++
...on Activity 7.1_ Managing Files and Creating Roles in Ansible_Interno.pdf | Bin 0 -> 1223407 bytes
HOA7.1_Managing_Files_and_Roles/README.md              | 1 +
HOA7.1_Managing_Files_and_Roles/files/default_site.html | 9 +++
HOA7.1_Managing_Files_and_Roles/inventory.yml           | 19 ++++++
HOA7.1_Managing_Files_and_Roles/roles/.gitkeep         | 5 ++
HOA7.1_Managing_Files_and_Roles/roles/base/main.yml    | 10 +++
HOA7.1_Managing_Files_and_Roles/roles/db_servers/main.yml | 11 ++++
HOA7.1_Managing_Files_and_Roles/roles/file_servers/main.yml | 4 ++
HOA7.1_Managing_Files_and_Roles/roles/web_servers/main.yml | 30 ++++++++
HOA7.1_Managing_Files_and_Roles/roles/workstations/main.yml | 12 ++++
HOA7.1_Managing_Files_and_Roles/site.yml               | 88 ++++++++
... Manage Enterprise Availability Monitoring via Ansible_Interno (1) (1).pdf | Bin 0 -> 806895 bytes
HOA8.1_Availability_Monitoring/README.md               | 1 +
HOA8.1_Availability_Monitoring/inventory.yml           | 10 +++
HOA8.1_Availability_Monitoring/roles/nagios/tasks/main.yml | 37 ++++++++
HOA8.1_Availability_Monitoring/site.yml               | 6 ++
Midterm_Skills_Exam/README.md                         | 1 +

(base) jerald@workstation:~/my-repo$ git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 6 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.13 KiB | 1.13 MiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/JeraldSanInterno/CPE232_Jerald_Interno.git
60779ad..1f56b11 main -> main
branch 'main' set up to track 'origin/main'.
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:

Answer the following:

1. What are the benefits of implementing containerizations?

- The benefits of the containerization serve as an application with all its dependencies so it runs consistently, quickly, and securely on any machine without setup hassles.

Conclusion

- The lab activity successfully demonstrated how to set up the Docker, automate image creation using ansible playbook, and organize the project within a structured git repository pushed to the GitHub. This workflow showcases how the containerization and infrastructure-as-code streamline modern software deployment, making applications consistent, reproducible, and easy to manage.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.



**COMPUTER
ENGINEERING**

QUEZON CITY

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.